

Jyothi Swaroop Ganapavarapu

AI Engineer | RAG Systems · Agentic AI · LLM Fine-tuning · MCP Engineering

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SUMMARY

AI Engineer with an MS in Data Science (**GPA 3.9, UNT, May 2026**) and **3+ years** of hands-on experience building and shipping Generative AI systems — specializing in **RAG pipelines, agentic AI, LLM fine-tuning, and Model Context Protocol (MCP) engineering**. Published researcher in *Science of the Total Environment* (**Elsevier, IF 8.2**) — one of very few industry AI engineers with peer-reviewed publication credit. Shipped **production-deployed AI systems** with live demos: hybrid RAG platform, multi-agent benchmarking suite, QLoRA fine-tuned SLM (**90% cost reduction**), and custom MCP server with persistent multi-turn memory — architecture-level decisions validated by benchmarks, not just wrappers.

SKILLS

- **AI / LLM Frameworks:** LangChain, LangGraph, HuggingFace PEFT, Groq, OpenAI API, Transformers, PyTorch
- **RAG & Retrieval:** FAISS, ChromaDB, BM25, BGE-large, GraphRAG, RRF Fusion, Cross-encoder Reranking, HNSW Indexing
- **Agentic AI:** ReAct, Plan-Execute, Tool Calling, Multi-Agent Orchestration, LangGraph State Machines, Model Context Protocol (MCP)
- **LLM Fine-tuning:** QLoRA, LoRA, 4-bit NF4 Quantization, Qwen3-4B, Llama 3.1, Mistral, Gemini 2.5 Flash
- **Evaluation & Benchmarking:** RAGAS (18-metric), ROUGE-L, BLEU, Tool-Call Accuracy, Latency Profiling, Hallucination Rate
- **Prompt Engineering:** Chain-of-Thought, Few-Shot, RAG Prompting, System Prompt Optimization
- **Deployment & MLOps:** FastAPI, Streamlit, HuggingFace Spaces, Azure, Docker, GitHub Actions, Selenium
- **Languages & Data:** Python, SQL, Pandas, NumPy, Scikit-learn, Plotly

EXPERIENCE

- Graduate AI/ML Student Assistant** Aug 2024 – May 2026
University of North Texas Denton, TX
 - Designed and deployed an **agentic AI analytics platform** using LangGraph state machines, tool-calling workflows, and structured prompt orchestration — enabling **15+ staff** to query institutional survey data in natural language, replacing manual BI/SQL analysis and reducing reporting time by **60%**.
 - Engineered a production **RAG pipeline** with semantic vector retrieval, metadata filtering, and context-aware prompting across **500+ survey records**; enforced JSON schema validation and output checks, achieving **near-zero hallucination rate** on domain-specific institutional queries.
 - Built autonomous report-generation workflows and scalable embedding/ingestion pipelines supporting **continuous indexing of 3+ department datasets**; benchmarked retrieval and factual accuracy, delivering production-ready insights for **5+ recurring department reviews**.
- AI Engineer** Jul 2022 – May 2024
Ofofo Pvt. Ltd. Bengaluru, India
 - Built **ML-powered vendor matching and recommendation pipelines** connecting SMBs with cybersecurity products across **100+ marketplace listings** — combining semantic similarity, TF-IDF ranking, and NLP-based tagging to improve product discovery accuracy by **~35%** (measured via match relevance scoring and client engagement metrics).
 - Designed and shipped an **agentic AI workflow** using LangChain and OpenAI APIs to automate cybersecurity questionnaire responses across **50+ B2B client engagements** — reducing per-questionnaire turnaround by **~40%** and enabling vendor due diligence with minimal manual effort.
 - Implemented a **ReAct-based AI agent** with tool-calling for compliance document retrieval and policy summarization across **SOC 2 and ISO 27001** audit workflows — reducing manual audit prep time by **~50%** and achieving **90%+** response accuracy across **200+ compliance queries** through iterative prompt optimization.
 - Engineered **data ingestion and preprocessing pipelines** in Python for structuring vendor, product, and transaction data across **3+ datasets**; automated recurring reporting workflows, cutting report generation from **~4 hrs to under 30 min** and reducing manual processing effort by **40%** across business review cycles.

PROJECTS

- Enterprise Agentic RAG Platform

LangGraph · LangChain · Groq · ChromaDB · BGE-large · BM25 · GraphRAG · Tavily · FastAPI · Streamlit

Live Demo | GitHub

Jan 2025 – Apr 2025

- Engineered a production-grade RAG platform using a **20-node LangGraph state machine** with **4-signal hybrid retrieval** — dense BGE-large vectors, BM25 sparse search, GraphRAG entity traversal, and temporal scoring — fused via Reciprocal Rank Fusion (RRF) with cross-encoder reranking across **481 indexed document chunks**.
 - Constructed a **183-entity, 177-relationship knowledge graph** via LLM extraction enabling graph-augmented retrieval; recording **BM25 fire rate 90%** and avg **2.8/4 retrieval signals** per query, validating multi-signal robustness.
 - Evaluated across **18 RAGAS metrics** on 10 benchmark queries — achieved **answer relevancy 1.00, 0% hallucination rate, 100% response success rate**, avg 3.6 citations/response, and 17/20 pipeline nodes executing per query. Deployed live on HuggingFace Spaces.
- Multimodal Agentic Assistant — ReAct vs Plan-Execute Benchmark

LangGraph · LangChain · Groq (Llama 3.3-70B / 8B / Mixtral) · Tavily · E2B · FastAPI · Streamlit · Plotly

Live Demo | GitHub

Feb 2025 – Apr 2025

- Architected a Multimodal Agentic Assistant implementing two agent paradigms — **ReAct** (Thought→Act→Observe loop) and **Plan-Execute** (Planner→Executor→Synthesizer nodes) — benchmarked head-to-head across **50 queries spanning 5 task categories**.
 - Integrated **5 heterogeneous tools** — Tavily web search, E2B sandboxed Python execution, Groq LLaMA Vision (image analysis), AST-based calculator, and Wikipedia search — enabling true multimodal capability across text, code, and image modalities.
 - Developed a custom benchmark harness computing ROUGE-L, tool-call accuracy, and latency across 3 Groq models; **ReAct achieved 87.3% tool-call accuracy vs 71.2% for Plan-Execute with 34% lower average latency**. Launched Streamlit dashboard with Plotly analytics.
- Low-Cost Resume Optimization via LLM Distillation into a Fine-Tuned SLM

Qwen3-4B · QLoRA (r=16, 4-bit NF4) · Gemini 2.5 Flash · HuggingFace PEFT · Selenium · FastAPI · Python

Live Demo | GitHub

Mar 2025 – May 2025

- Designed an end-to-end **LLM knowledge distillation pipeline** — scraped **249 job descriptions** and extracted **1,800+ resumes** as domain corpus; used Gemini 2.5 Flash Batch API as teacher model to generate **1,530 high-quality (instruction, output) training pairs**.
 - Fine-tuned Qwen3-4B using **QLoRA** (rank=16, alpha=32, 4-bit NF4 quantization), reducing VRAM footprint from **28GB to under 8GB** while maintaining task quality; delivering **90% cost reduction vs GPT-4** inference with **<120ms latency** on domain-specific resume optimization tasks.
 - Validated distillation pipeline on held-out test pairs achieving **ROUGE-L score of 0.74** vs teacher model output, confirming task-quality parity from a **4B-parameter student model** at **90% lower inference cost**; shipped end-to-end pipeline via live FastAPI frontend demo.
- MCP-Powered AI Research Assistant

fastmcp · LangGraph · LangChain · Groq (LLaMA 3.3-70B) · arXiv API · Tavily · Streamlit

Live Demo | GitHub

Apr 2025 – May 2025

- Designed and built a custom **MCP (Model Context Protocol) server** using fastmcp exposing **5 standardized research tool endpoints** — `search_arxiv`, `web_search`, `summarize_paper`, `format_citation`, and `compare_papers` — with `format_citation` at **<5ms** and `search_arxiv` at **sub-800ms** latency.
 - Engineered a LangGraph ReAct agent as the MCP client connecting to the server at runtime via the MCP protocol; agent dynamically chains tools across multi-step research pipelines (`search` → `summarize` → `compare` → `cite`) averaging **9 tool calls per query** at **36.5s end-to-end latency**; powered by Groq LLaMA 3.3-70B delivering structured **5-section comparisons** and **3–5 sentence summaries**.
 - Implemented **persistent multi-turn memory** with JSON-backed session storage enabling cross-session research continuity; agent generates APA and BibTeX citations on demand. Deployed live on HuggingFace Spaces (Groq, Tavily, arXiv).

PUBLICATIONS

- A Novel Seaweed Detection Image Processing and Validation Framework:
A Pragmatic Study on Natural Seaweed Beds along North-West Coast of India

Science of the Total Environment, Elsevier · Vol. 978 · Impact Factor: 8.2 · 10.1016/j.scitotenv.2025.179296

May 2025

EDUCATION

- University of North Texas

Master of Science in Data Science GPA: 3.9/4.0

Aug 2024 – May 2026

Denton, Texas
- KL University

B.Tech in ECE — Specialization: Data Science & Big Data Analytics CGPA: 9.53/10

Jul 2020 – May 2024

Vijayawada, India

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Microsoft Certified: Azure Data Fundamentals (DP-900)
- Microsoft Certified: Azure Fundamentals (AZ-900)